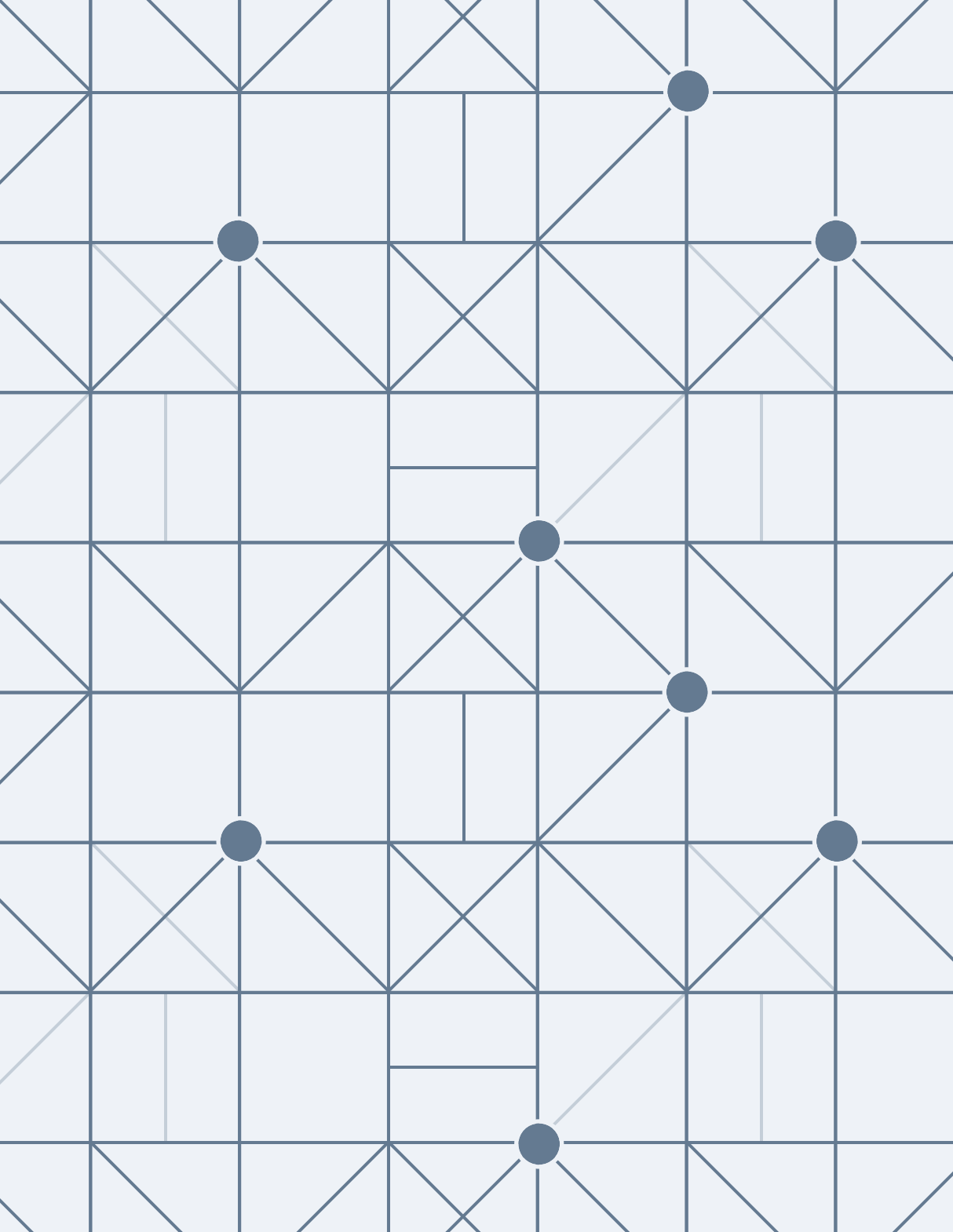




NOTES on SIMPLEX METHOD

THE SIMPLEX METHOD CONSIDERS ONLY CORNER-POINT FEASIBLE (CPF) SOLUTIONS AND MOVES FROM ONE TO ANOTHER THAT IS ADJACENT (i.e., THEY SHARE $n-1$ CONSTRAINTS)

ADJECENCY IS USEFUL FOR OPTIMALITY: IF A CPF SOL HAS NO ADJACENT CPF SOL THAT GIVES A BETTER OBJECTIVE FUNCTION, THEN IT MUST BE OPTIMAL.

ALGEBRAIC PROCEDURE

BASED ON SOLVING A SYSTEM OF EQUATIONS

=> CONVERT INEQUALITY CONSTRAINTS TO EQUALITY CONSTRAINTS WITH SLACK VARIABLES (AUGMENTED FORM)



- IF A SLACK VARIABLE $= 0$ IN A SOLUTION THEN THIS SOLUTION LIES ON THE CONSTRAINT BOUNDARY OF THE CORRESPONDING CONSTRAINT
- IF A SLACK VARIABLE < 0 IN A SOLUTION THEN THIS SOLUTION LIES ON THE INFEASIBLE SIDE OF THE CORRESPONDING CONSTRAINT BOUNDARY

REMARK: $\# \text{VARIABLES} - \# \text{EQUATION} = \text{DEGREES OF FREEDOM (d)}$

=> d VARIABLES (NON-BASIC VARIABLES) CAN BE ARBITRARILY SET TO ZERO TO SOLVE THE EQUATIONS IN THE REMAINING VARIABLES (BASIC VARIABLES) LEADING TO A BASIC SOLUTION

=> ADJACENT BF SOL HAVE ALL BUT ONE NON-BASIC VAR IN COMMON. EQUIVALENTLY THEY HAVE ALL BUT ONE BASIC VARIABLE IN COMMON (NOT NECESSARILY WITH THE SAME VALUE)

INITIALIZATION : CHOOSE NON-BASIC VARIABLES TO BE $= 0$
[IF POSSIBLE START FROM ORIGIN]

SOLVE THE SYSTEM FOR THE (REMAINING) BASIC VARIABLES
[READ VALUES OF THE RIGHT-HAND-SIDE]

OPTIMACITY TEST : IF NO NON-BASIC VARIABLE CAN PRODUCE AN INCREASE IN THE OBJECTIVE FUNCTION (BY GOING FROM 0 TO A FEASIBLE VALUE) WE ARE DONE!

IF THE COEFFICIENTS OF THE OBJECTIVE FUNCTION EQUATION, CORRESPONDING TO THE NON-BASIC VARIABLES ARE NEGATIVE (I.E. THEY CAN PRODUCE A POSITIVE INCREASE), THEN THE BF SOL IS NOT OPTIMAL

ITERATION : 1. DIRECTION OF MOVEMENT
INCREASE ONE NON-BASIC VARIABLE FROM ZERO (I.E. MOVE ALONG ONE EDGE DEPARTING FROM THE CURRENT CPF SOL)

=> ENTERING BASIC VARIABLE IS THE ONE THAT PRODUCE THE HIGHEST INCREASE (I.E. THE ONE WITH THE MOST NEGATIVE COEFFICIENT IN THE OBJECTIVE FUNCTION EQUATION)

PIVOT
COLUMN

2. WHEN TO STOP

INCREASE THE EBV AS MUCH AS POSSIBLE AS LONG AS THE CONSTRAINTS ARE SATISFIED. NOTE THAT INCREASING THE VALUE OF THE EBV CHANGES THE VALUES OF THE OTHER BV (WHICH HAS TO BE NON-NEGATIVE)

=> CHECK THE EFFECT OF THE EBV ON THE OTHER BV AND STOP AS SOON AS THE "FIRST" BV DROP TO ZERO

[THE MINIMUM RATIO TEST GIVES THE SMALLEST VALUE THE EBV CAN TAKE WITHOUT VIOLATING THE CONSTRAINTS]

Pivot
Row

=> LEAVING BASIC VARIABLE IS THE ONE THAT DROP TO ZERO (i.e. to non-basic) WHEN THE EBV INCREASES

3. NEW BF SOLUTION

CONVERT THE SYSTEM OF EQUATIONS TO A MORE CONVENIENT FORM BY GAUSSIAN ELIMINATION TO CONDUCT THE OPTIMALITY TEST AND COMPUTE THE NEW BF SOL

[NON-BASIC VARIABLES ARE $= 0$ AND BASIC VARIABLES CAN BE READ ON THE RIGHT-HAND-SIDE]

4. OPTIMALITY TEST

[SEE ABOVE]

